Variant Interpretation in CYP2C19

BIOT 6900 — Module 5, Computational Lab 5
Mohammed Sharukh
Gene: CYP2C19 (self-selected; no allocation received) · AlphaGenome Atlas: run live

6.1 · Thresholds and calibration

Frequency threshold

I set a maximum credible allele frequency of 0.50 for CYP2C19. This removed 2 of 80 variants and, critically, removed no P/LP or drug-response variants. A rare-disease threshold of 0.001 assumes that variants with a high frequency in the population are less likely to have a harmful effect. However, CYP2C19*2 and *17 are common variants that can still affect how a person metabolizes clopidogrel, so using this strict cutoff would incorrectly remove important pharmacogenomic variants.

For contrast, a 0.001 filter would have discarded 19 variants, including both known star alleles for this gene: CYP2C19*2 (rs4244285, gnomAD AF 0.168) and CYP2C19*17 (rs12248560, AF 0.200).

For rare-disease genes, variants that cause severe diseases early in life are more likely to be removed by natural selection, which is why they usually remain rare. However, CYP2C19 variants like *2 and *17 affect how a person responds to certain drugs, so they do not face the same selection pressure and can be common while still having important clinical effects.

Choice of predictor

I chose CADD because it covered more of the CYP2C19 variants, so I could calibrate the predictor using a larger and more representative set of variants. REVEL could not score as many variants, and leaving those variants out would reduce the amount of data available for calibration and might cause me to miss important variants in some consequence classes.

Coverage in my audit: CADD 85.0% (68 of 80), REVEL 67.5% (54 of 80), AlphaMissense 67.5% (54 of 80). REVEL returned no score at all in seven consequence classes.

If a tool cannot score a variant, it does not mean the variant is harmless; it simply means the tool cannot predict its effect. For example, REVEL could not score rs72558186, the splice donor variant, so I cannot use that blank result as evidence that the variant is benign.

Calibration result

The calibration returned NaN for all likelihood ratios because there were zero P/LP and zero B/LB labelled variants available to calculate them. Therefore, the likelihood ratios were undefined, and none of the score bands had enough labelled data to establish reliable evidence. This shows that the ACMG framework cannot be directly applied to CYP2C19 pharmacogenomic variants without appropriate pathogenicity labels, because drug-response variants may have important clinical effects without being classified as pathogenic or benign.

ClinVar significance breakdown for the 78 variants after filtering:

ClinVar significance	n
drug response	37
Uncertain significance	33
Likely benign	3
Benign	3

ClinVar significance	n
Benign; other	1
other	1
Pathogenic / Likely pathogenic	0

Calibration table (cadd_phred, bands [0, 10, 20, 25, 30, 50]): every band returned n = 0, n_PLP = 0, n_BLB = 0, LR+ = NaN, evidence = undefined.

I cannot confidently use the predictor scores to classify variants because the calibration returned undefined likelihood ratios, with zero P/LP and zero B/LB labelled variants. Out of the 78 variants, 37 were labelled as drug response and 33 as uncertain significance, so there was not enough suitable data to calculate reliable likelihood ratios. This is why I kept all 33 VUS unchanged in Q5 instead of reclassifying them based on scores alone.

6.2 · Coverage gaps and the regulatory genome

Which classes no missense predictor could score

Seven consequence classes returned no REVEL score at all: synonymous_variant, upstream_gene_variant, intergenic_region, intron_variant, splice_donor_variant, stop_lost and stop_gained. The coverage audit was as follows.

Consequence	n	CADD	REVEL	AlphaMissense
missense_variant	54	53	53	54
synonymous_variant	7	6	0	0
upstream_gene_variant	5	4	0	0
intergenic_region	1	1	0	0
intron_variant	1	1	0	0
start_lost	1	1	1	0
splice_donor_variant	1	0	0	0
stop_lost	1	1	0	0
stop_gained	1	1	0	0

One specific unscored variant

rs72558186, a splice donor variant, had no CADD, no REVEL and no AlphaMissense score — the only variant in the gene that no conventional tool could reach.

For rs72558186, I would look for evidence of abnormal mRNA splicing, such as exon skipping or intron retention, because a splice donor variant can affect how the mRNA is processed. I would use an RNA sequencing assay or RT-PCR followed by sequencing to compare the splicing pattern with a normal control and determine whether the variant disrupts normal CYP2C19 mRNA processing.

I would use liver tissue or hepatocytes because CYP2C19 is mainly expressed in the liver, where it helps metabolize drugs. A blood sample might not have enough CYP2C19 expression to clearly detect abnormal splicing, so liver cells would be a more suitable source for the experiment.

What the Atlas added

The Atlas added useful evidence for variants that the other tools could not score, especially rs72558186, which received an AVI of 25.51 with MERGED_SPLICING as the top feature. It also highlighted the CYP2C19*2 variant (rs4244285) with an AVI of 22.33 despite its low CADD score of 5.686. However, most attributions came from

AlphaMissense, so the main additional value was its ability to capture splicing and other regulatory evidence that traditional missense-focused tools could miss.

The Atlas scored 70 variants with 0 failures. Attribution breakdown:

top_feature	n	What it means
ALPHAMISSENSE	124	coding mechanism — missense tools already saw these
CACTUS_241_WAY	30	conservation carrying the score; mechanism NOT resolved
MERGED_SPLICING	7	splicing mechanism — an RNA assay would settle it
PHASTCONS_470_WAY	4	conservation; mechanism NOT resolved
MAX_ABS_DNASE	2	regulatory mechanism, traceable to a tissue
START_LOST / PROTEIN_TERMINATION / STOP_LOST	1 each	coding consequence

Most of the Atlas attributions came from ALPHAMISSENSE, with 124 out of approximately 170, while only 7 came from MERGED_SPLICING and 2 from MAX_ABS_DNASE. This suggests that much of the Atlas output overlapped with evidence already captured by missense prediction tools. Another 30 attributions came from CACTUS_241_WAY, which mainly reflects evolutionary conservation rather than explaining exactly how a variant affects gene function. The Atlas was useful for identifying some additional signals, but not every high score gave a clear explanation of the variant's effect.

6.3 · A discordant variant, read mechanistically

I chose rs4244285 (CYP2C19*2). It tripped flag_atlas_only: REVEL and AlphaMissense returned no score at all because the variant is annotated synonymous, and CADD scored it very low (2.606–5.686 across the two records for this position), while the Atlas was the only tool to flag it, assigning an AVI of 22.33 with MERGED_SPLICING as the top feature.

I think both outputs can be correct because CYP2C19*2 is a synonymous variant, meaning it does not change the amino acid sequence, so CADD may assign it a low score based on its predicted protein-level impact. However, the variant disrupts normal mRNA splicing, which can affect the production of a functional protein. This explains why the Atlas gives it a high AVI score and identifies MERGED_SPLICING as the top feature, showing that a variant can have a major functional effect without changing the protein's amino acid sequence directly.

CYP2C19*2 is called a synonymous variant because the nucleotide change does not directly change the amino acid at that position. However, it creates a cryptic splice site in exon 5, which leads to abnormal mRNA splicing and a truncated, non-functional protein. This means that even though the change looks harmless at the amino acid level, it can still stop the gene from producing a functional enzyme.

CYP2C19*2 is a common variant, with an allele frequency of around 16.8% in my dataset, and it is known to reduce CYP2C19 enzyme function and affect how a person responds to clopidogrel. Its effect is important because CYP2C19 testing can help guide clopidogrel treatment. CADD gave it a low score, showing that a predictor can underestimate an important variant when its main effect is on mRNA splicing rather than directly changing the protein's amino acid sequence.

Discordance summary for CYP2C19: 70 variants tripped at least one rule — flag_no_tool_at_all 58, flag_atlas_only 21, flag_revel_vs_am 6, flag_cadd_vs_revel 3, flag_scored_but_vus 0, flag_common_but_plp 0. Fourteen of the flagged variants carried an rsid.

6.4 · What I could not determine

My analysis cannot establish zygosity and phase, meaning whether a patient carries one or two copies of CYP2C19*2 and whether it occurs alongside other relevant variants on the same chromosome. It also cannot

establish clinical utility, because predictor scores and allele frequencies alone cannot tell us how well using this information to guide clopidogrel treatment improves outcomes for an individual patient.

Zygosity matters because a person with one copy of CYP2C19*2 may be an intermediate metaboliser, while someone with two copies may be a poor metaboliser. These differences can affect clopidogrel treatment recommendations. My dataset only lists individual variants, so I cannot tell how many copies a patient carries or determine their actual metaboliser status from the variant list alone.

The experiment that would resolve it

I would use an RT-PCR assay followed by sequencing to compare CYP2C19 mRNA splicing in liver cells carrying *2 with a normal control. I would expect to see an aberrant transcript caused by the cryptic splice site; if the assay showed normal splicing with no abnormal transcript, that would make me reconsider my interpretation that *2 causes loss of function through disrupted splicing.

I would also look for a clinical pharmacokinetics dataset comparing clopidogrel active-metabolite levels in people who carry CYP2C19*2 with those who do not. If people carrying *2 consistently had normal active-metabolite levels, it would make me question my interpretation and investigate whether other factors were affecting the results. This would help connect the predicted molecular effect to what actually happens in patients.

Addendum: Gene and data choice; what the data let me claim

BIOT 6900 · Module 5 · Mohammed Sharukh. Added October 29, 2026, per the instructor's announcement asking every report to include a data-choice section with biological interpretation and a limitations section. The analysis above is unchanged; the discordant variant (6.3) and limits (6.4) are already covered.

A. Gene and data choice, with biological interpretation

Gene. No gene allocation was available on Canvas, so I chose CYP2C19 from the course list (CYP2C19, CYP2C9, VKORC1, DPYD, TPMT). None of these five is a PDAC gene, so this module necessarily departs from the PDAC thread of my other assignments. CYP2C19 is a hepatic cytochrome P450 that converts the prodrug clopidogrel into its active form and metabolizes proton-pump inhibitors and several antidepressants. That makes its variants clinically actionable even though they do not cause disease.

Data. I fetched 80 ClinVar variants through MyVariant.info, with CADD (hg19), REVEL, AlphaMissense and gnomAD allele frequencies, and scored 70 variants with the AlphaGenome Atlas (0 failures). All sources are open; the Atlas needs a free non-commercial API key, kept in an environment variable.

Known variants recovered. Both anchor star alleles were present: CYP2C19*2 (rs4244285, AF 0.168), the main loss-of-function allele, and CYP2C19*17 (rs12248560, AF 0.200), an increased-function promoter allele. This confirms the fetch was complete. Their high frequencies are themselves the biological lesson: drug-response alleles escape the selection that keeps rare-disease variants rare, so a 0.001 filter would have discarded both.

Surprising results. (1) *2, the most clinically important variant, looks harmless to the standard tools: it is annotated synonymous, REVEL and AlphaMissense give it no score, and CADD gives it 5.686. Only the Atlas flagged it (AVI 22.33, MERGED_SPLICING), consistent with its known mechanism: a cryptic splice site that truncates the protein. (2) Calibration was impossible: 0 P/LP and 0 B/LB labels after filtering, because ClinVar labels pharmacogene variants "drug response" rather than pathogenic or benign.

B. What these data let me claim, and not claim

What I can claim. I can claim which variant classes each tool can and cannot see (seven consequence classes had no REVEL score; the splice-donor variant rs72558186 had no conventional score at all), that a pharmacogene needs a frequency threshold far above rare-disease practice, and that the Atlas adds a splicing signal missense tools miss.

What I cannot claim. I cannot classify any VUS: with undefined likelihood ratios all 33 remain VUS, and moving them on raw scores would be unsupported. I cannot infer a patient's metaboliser status, which depends on zygosity and phase; equate a high predictor score with pathogenicity; or establish clinical utility. A label-parsing issue (`clinvar_label()` returned "none" for 7 benign-side variants) does not change this conclusion, because the P/LP side stays at zero.